

how do you calculate the grundy number?

Asked 4 years, 8 months ago Modified 4 years, 8 months ago Viewed 1k times

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I want to calculate the grundy number of these 4 normal-play nim heaps: $0, 7, 7, 7$ I'm confused when comparing two wikipedia pages: [sprague-grundy](#) suggests I should find the mex of these 4 numbers, which is 1 . while the [nim](#) page says I should XOR them all, which generates 7 .

which is it?

[game-theory](#) [combinatorial-game-theory](#)

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asked May 11, 2019 at 12:57

 **ihadanny**
181 9

1 Answer

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The Grundy number is the nim-sum (i.e. the xor) of the numbers of items in all the heaps—namely 7 in this case.

You have misunderstood what the Wikipedia page is saying. There's no shame in that, though, since it's not particularly well written. The numbers $n_1, n_2, \dots, n_k$, over which the page says to take the *mex* aren't numbers of items in nim heaps, but the Grundy numbers of *the positions you can move to* in the game G' . If G' were your game of nim with three heaps of size seven, for instance, there are effectively seven positions you can move to—two heaps of size seven, and one heap of any size from zero to six. The Grundy numbers of these positions are $0, 1, 2, 3, 4, 5, 6$, so the *mex* of these numbers, namely 7 , is the Grundy number of the position you're facing.

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answered May 11, 2019 at 14:03

 **lonza leggiera**
28.1k 2 12 33

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thanks! where did you see the "positions you can move to" bit? can you recommend a better source for learning about the grundy theorem and solving nim?

– ihadanny May 11, 2019 at 16:48